Pro-Range PGM45775-50.9K Planetary Gear DC Motor with ME-775 Encoder (7PPR)

Datasheet Summary

Model: PGM45775-50.9K
Motor Type: 775 Brushed DC Motor with Planetary Gearbox
Operating Voltage: 12V DC
Rated Speed: 92 RPM
Rated Torque: 234.7 N·cm (23.93 kg·cm)
Gear Ratio: 50.9:1
Rated Current: 4.02 A
Rated Power: 47.28 W
Base Motor Speed: 6000 RPM
Shaft Diameter: 10 mm
Shaft Length: 25 mm
Shaft Type: Keyed Shaft
Motor Diameter: 45 mm
Motor Length: 125 mm
Weight: 880–985 g
Operating Temperature: -40°C to 100°C

Encoder Specifications

Encoder Model: ME-775
Encoder Type: Hall-Effect Quadrature Encoder
Resolution: 7 Pulses Per Revolution (PPR) at Motor Shaft
Outputs: Channel A and Channel B
Applications: Speed Measurement, Direction Detection, Position Feedback

Encoder Pinout

1 - Encoder VCC
2 - Encoder GND
3 - Channel A
4 - Channel B
5 - Motor -
6 - Motor +

CPR Calculation After Gearbox

Motor Encoder Resolution = 7 PPR
Gear Ratio = 50.9:1

Output Shaft Pulses/Revolution = 7 × 50.9 = 356.3 ≈ 356 CPR

Quadrature Counting:
• 1x Counting: ≈ 356 CPR
• 2x Counting: ≈ 713 CPR
• 4x Counting: ≈ 1425 CPR

At Rated Speed (92 RPM):
• 1x Counting: ≈ 546 pulses/sec
• 4x Counting: ≈ 2185 counts/sec

Typical Applications

• AGV and Mobile Robots
• Conveyor Systems
• Robotic Arms
• Automation Equipment
• Position-Control Systems
• Differential Drive Robots